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A Project Report on

"ATTENDNET: A REAL-TIME DEEP LEARNING FRAMEWORK FOR MULTI-FACE RECOGNITION AND AUTOMATED ATTENDANCE MONITORING"

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Submitted by

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CERTIFICATE

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ABSTRACT

Automated attendance monitoring in educational institutions has traditionally relied on manual methods, which are time-consuming, error-prone, and susceptible to proxy attendance. Although biometric systems such as fingerprint scanners provide automation, they require specialized hardware and physical contact, making them less scalable and less convenient for large classrooms. To address these challenges, this paper presents **AttendNet AI**, a temporal-aware real-time face recognition system designed for automated attendance monitoring in dynamic classroom environments.

The system utilizes a pre-trained deep learning model based on metric learning to generate 128-dimensional facial embeddings for each individual. These embeddings act as unique feature representations and are compared with stored data using Euclidean distance to accurately identify users. A key innovation of the system is the incorporation of a multi-frame temporal validation mechanism, which ensures that attendance is marked only after consistent recognition across multiple consecutive frames. This approach significantly reduces false positives caused by temporary disturbances such as motion blur, partial occlusion, or brief face appearances.

AttendNet AI is implemented as a full-stack web application and deployed online, making it easily accessible through a browser. The system includes four main portals: administrator authentication for secure access, student registration for enrollment, a real-time attendance dashboard for monitoring, and a student portal for viewing attendance history.

The system is evaluated under real classroom conditions with 90 registered individuals and achieves an accuracy of 96.3%, precision of 96.9%, recall of 95.0%, and an F1-score of 95.6%. These results demonstrate the effectiveness and reliability of the proposed approach. Additionally, the system operates efficiently on standard CPU-based hardware without requiring GPU acceleration, making it cost-effective and suitable for deployment in resource-constrained educational institutions.

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CHAPTER 1

1.1 INTRODUCTION

Attendance monitoring is an essential administrative activity in educational institutions, as it helps track student participation, discipline, and academic performance. Traditional methods such as manual roll calls are time-consuming and prone to errors, especially in large classrooms. These methods are also vulnerable to proxy attendance, which reduces their reliability. To overcome these issues, automated systems like fingerprint scanners and RFID cards have been introduced, but they require specialized hardware, physical interaction, and are not efficient for handling multiple students simultaneously in real time.

With advancements in Artificial Intelligence, particularly in Computer Vision and Deep Learning, face recognition has become a powerful and contactless solution for attendance management. It works by extracting unique facial features and matching them with stored data to identify individuals. This approach eliminates the need for physical contact and additional hardware, making it suitable for modern smart classrooms.

However, many existing face recognition attendance systems rely on single-frame detection, which is not reliable in real-world scenarios. Factors such as lighting variations, facial movements, occlusions, and camera noise can affect accuracy, leading to incorrect or missed attendance marking. These limitations highlight the need for a more robust and intelligent system.

To address these challenges, this project introduces **AttendNet AI**, a real-time face recognition attendance system with temporal awareness. Unlike traditional systems, it uses a multi-frame temporal validation mechanism to confirm identity across consecutive frames, improving accuracy and reducing false detections. The system is built using dlib-based 128-dimensional facial embeddings and supports real-time multi-face detection.

In addition, AttendNet AI is developed as a full-stack web application with separate interfaces for administrators and students, along with a real-time attendance dashboard. It is deployed online and can be accessed through a web browser, making it scalable, user-friendly, and suitable for real-world implementation without requiring specialized hardware.

1.2 MOTIVATION BEHIND PROJECT TOPIC

The motivation for this project comes from the need to improve the efficiency and reliability of attendance systems in educational institutions. Manual methods are slow and prone to misuse, while

hardware-based biometric systems are costly and not scalable. There is a need for a solution that is automated, accurate, and easy to use without requiring additional devices.

Face recognition provides a promising alternative, but existing systems often fail in real-world conditions due to reliance on single-frame detection. This project is motivated by the goal of building a system that performs reliably under dynamic conditions such as lighting changes and subject movement. The use of temporal validation helps improve accuracy and ensures consistent performance. Additionally, the project aims to apply concepts of artificial intelligence and full-stack development to create a practical and deployable solution.

1.3 AIM AND OBJECTIVES OF THE WORK

Aim

The aim of this project is to develop a real-time, accurate, and contactless attendance monitoring system using deep learning-based face recognition techniques integrated with a temporal validation mechanism. The system is designed to operate efficiently on standard hardware without the need for specialized devices, making it suitable for practical deployment in educational institutions. By leveraging computer vision and artificial intelligence, the proposed system ensures reliable identification of individuals while minimizing errors caused by environmental variations such as lighting, movement, and occlusion.

Objectives

- The primary objective of this project is to design and implement a robust face recognition-based attendance system capable of detecting and recognizing multiple faces simultaneously in a live video stream. The system aims to handle real-time processing efficiently while maintaining high accuracy and performance.
- Another important objective is to develop a temporal validation mechanism that verifies the identity of individuals across multiple consecutive frames before marking attendance. This approach improves reliability by reducing false positives and ensuring consistent recognition even in dynamic conditions.
- The project also focuses on studying various data acquisition techniques and understanding the role of pre-trained deep learning models in face recognition. This includes exploring methods for collecting, preprocessing, and transforming facial data into meaningful embeddings for accurate comparison.

CHAPTER 2

Literature Survey

2.1 Early Face Recognition Methods

Early face recognition techniques were primarily based on appearance-driven approaches and statistical methods for feature extraction. One of the most influential techniques was the Eigenfaces method proposed by Turk and Pentland, which utilized Principal Component Analysis (PCA) to reduce high-dimensional facial data into a lower-dimensional representation while preserving the most significant variance in the dataset [2]. This approach transformed facial images into a set of orthogonal basis vectors, enabling efficient comparison and recognition. However, Eigenfaces suffered from limitations such as sensitivity to lighting variations, pose differences, and facial expressions.

To address these issues, Fisherfaces were introduced by Belhumeur et al., which applied Linear Discriminant Analysis (LDA) to improve class separability by maximizing the ratio of between-class variance to within-class variance [3]. This method provided better discrimination between individuals compared to PCA-based methods. Despite these improvements, both Eigenfaces and Fisherfaces were still affected by environmental factors such as illumination changes and occlusion, limiting their practical applicability in real-world scenarios.

To enhance robustness, feature-based methods such as Scale-Invariant Feature Transform (SIFT) and Histogram of Oriented Gradients (HOG) were developed. SIFT focuses on detecting and describing local key points that are invariant to scale and rotation, making it more robust to variations in image conditions [4]. Similarly, HOG extracts gradient orientation features that capture edge and shape information, which are particularly useful for object detection tasks [5]. Among these, HOG became widely adopted for face detection due to its balance between computational efficiency and accuracy.

These early techniques laid the foundation for modern face recognition systems by introducing fundamental concepts such as dimensionality reduction, feature extraction, and pattern matching. However, their inability to handle complex variations in real-world environments highlighted the need for more advanced approaches. This evolution is well documented in comprehensive studies on face recognition methodologies [1], [20]. The HOG-based detection approach remains relevant today and is used in AttendNet AI for efficient real-time face detection on standard hardware.

2.2 Deep Learning-Based Face Recognition

The introduction of deep learning marked a significant turning point in the field of face recognition by enabling automatic feature extraction and significantly improving accuracy. DeepFace, developed by Taigman et al., demonstrated near human-level performance in face verification tasks by using deep

neural networks trained on large-scale datasets [6]. This breakthrough highlighted the potential of deep learning in overcoming the limitations of traditional methods.

Subsequently, FaceNet introduced a powerful embedding-based approach that uses triplet loss to learn a mapping from facial images to a high-dimensional feature space [7]. In this space, images of the same individual are placed closer together, while those of different individuals are positioned farther apart. This approach allows efficient comparison of faces using simple distance metrics such as Euclidean distance, eliminating the need for retraining when new identities are added.

Another important contribution is VGGFace, which demonstrated that deep convolutional neural networks trained on large datasets can generate highly discriminative facial embeddings [8]. These embeddings capture complex facial features and provide high accuracy even under varying conditions. Furthermore, advancements in deep architectures such as ResNet introduced skip connections, enabling the training of very deep networks without degradation in performance [9]. This significantly improved feature extraction capabilities and overall model accuracy.

The AttendNet AI system adopts a similar deep learning-based approach by utilizing a pre-trained model from the dlib library, which generates 128-dimensional facial embeddings [15]. This approach enables efficient recognition and supports dynamic user enrollment without retraining, making the system scalable and suitable for real-time applications. Additionally, the use of modern frameworks such as TensorFlow and OpenCV further supports efficient implementation and deployment [18], [21], [19]. These advancements have made deep learning the dominant approach in modern face recognition systems.

2.3 Real-Time Detection and System Limitations

With the advancement of deep learning, real-time face detection and recognition became feasible through the development of efficient object detection algorithms. One of the most notable contributions in this area is YOLO (You Only Look Once), which enables fast and accurate detection of multiple objects, including faces, in real-time video streams [13]. YOLO processes the entire image in a single pass, making it significantly faster compared to traditional detection methods.

Other deep learning-based detection techniques have also improved the ability to process dynamic visual data efficiently. These advancements have enabled the development of systems capable of handling real-time video streams and detecting multiple faces simultaneously. However, despite these improvements, many existing face recognition-based attendance systems still rely on single-frame recognition.

This reliance on single-frame processing introduces several limitations. In real-world environments,

factors such as motion blur, lighting variations, partial occlusions, and temporary visibility of faces can affect detection accuracy. As a result, systems may incorrectly identify individuals or fail to recognize them consistently. These issues are particularly prominent in dynamic environments such as classrooms, where students frequently move or change positions.

Moreover, single-frame systems are more susceptible to false positives, where a face is incorrectly recognized due to temporary resemblance or noise in the data. These limitations highlight the need for more robust approaches that consider temporal consistency across multiple frames rather than relying on isolated detections. Recent studies emphasize the importance of temporal information in improving recognition reliability in video-based systems [22].

2.4 Proposed System and Research Gap

The analysis of existing literature reveals a clear research gap in ensuring reliable and consistent face recognition in real-time environments. While traditional and deep learning-based methods have significantly improved recognition accuracy, most systems focus primarily on single-frame analysis and fail to address temporal inconsistencies. This limitation reduces system reliability in practical scenarios where environmental conditions are constantly changing.

The proposed AttendNet AI system addresses this gap by introducing a temporal validation mechanism that confirms identity across multiple consecutive frames before marking attendance. By incorporating temporal consistency, the system reduces false positives and ensures more reliable recognition. This approach enhances system stability and makes it more suitable for real-world applications.

Unlike many existing systems that require GPU acceleration or large-scale retraining, AttendNet AI is designed as a lightweight, CPU-based solution capable of real-time operation. It leverages pre-trained embedding models, enabling dynamic enrollment of users without retraining. Additionally, the system integrates a full-stack web application with role-based access control, providing features such as user registration, real-time attendance monitoring, and secure data management.

This combination of temporal validation, efficient embedding-based recognition, and web-based deployment makes AttendNet AI a practical and scalable solution for educational institutions. The system not only addresses the limitations of existing approaches but also demonstrates how modern technologies can be effectively integrated to solve real-world problems.

Sl.no	Paper Title	Author	Year	Methods Used
1	Eigenfaces for Recognition	Turk and Pentland	2021	PCA-based Feature Extraction
2	FaceNet: A Unified Embedding for Face Recognition	Hongen Lu et al.	2020	Triplet Loss, Embedding Learning
3	Deep Face Recognition (VGGFace)	Sammy V. Militante et al.	2021	Convolutional Neural Networks
4	Deep Residual Learning for Image Recognition	Nanette V. Dionisio et al.	2021	Deep Residual Networks (ResNet)

Table 2.1: Literature Survey Summary

CHAPTER 3

Software Requirement Specifications

3.1 Introduction

The Software Requirement Specification (SRS) document provides a detailed and structured description of the AttendNet AI system, which is designed for real-time face recognition-based attendance monitoring. This document outlines the overall functionality, system behavior, constraints, and requirements necessary for successful implementation. AttendNet AI integrates computer vision and deep learning techniques with a web-based platform to automate the attendance process efficiently. The use of facial embeddings along with a temporal validation mechanism enhances the accuracy and reliability of the system. This document serves as a reference guide for developers, testers, and users throughout the system development lifecycle.

3.1.1 Project Purpose

The main purpose of the AttendNet AI system is to develop a fully automated, contactless attendance solution that eliminates the limitations of traditional attendance methods. Manual attendance systems are time-consuming and prone to errors such as proxy attendance, while biometric systems require specialized hardware. AttendNet AI addresses these issues by using face recognition combined with temporal validation to ensure accurate identification. The system verifies a user's identity across multiple frames before marking attendance, thereby improving reliability. Additionally, it provides a centralized platform for storing, managing, and accessing attendance records efficiently.

3.1.2 Project Scope

The scope of AttendNet AI includes real-time detection and recognition of multiple faces using a webcam or video stream. The system supports dynamic enrollment, allowing new users to be added without retraining the model, which enhances scalability. It includes a web-based interface developed using Flask, where administrators can register users, manage attendance, and monitor system performance, while students can view their attendance records. The system is deployed online, making it accessible from any device with internet connectivity. This eliminates the need for specialized hardware and ensures ease of integration into existing institutional systems.

3.1.3 User Classes and Characteristics

AttendNet AI is designed to support two main categories of users: administrators and students. Administrators have full control over system operations, including user registration, attendance management, and monitoring system performance. Students have limited access, primarily to view their attendance records. The system is intended for users with basic computer literacy, ensuring that they can easily interact with the application without requiring technical expertise. The user interface is designed to be simple, intuitive, and user-friendly, enabling smooth interaction and quick adoption in educational environments.

3.1.4 Assumptions and Dependencies

The system assumes that the input video stream is clear and of sufficient quality to enable accurate face detection and recognition. It also assumes that users are registered with properly captured facial images during enrollment. The performance of the system depends on environmental factors such as lighting conditions, camera resolution, and processing capabilities of the hardware. AttendNet AI relies on several external libraries, including OpenCV for image processing, dlib for facial embedding extraction, and Flask for backend development. It also depends on a cloud-based database for storing user data and attendance records. A stable internet connection is required for accessing the deployed system.

3.2 Functional Requirements

The system provides essential functionalities required for automated attendance monitoring. It captures live video input and processes frames in real time to detect faces using HOG-based detection. Detected faces are converted into 128-dimensional embeddings using a pre-trained deep learning model. These embeddings are then compared with stored data to identify individuals accurately.

A key feature of the system is the temporal validation mechanism, which ensures that attendance is marked only after consistent recognition across multiple frames. This reduces false positives and improves system reliability. Additionally, the system supports user registration by capturing and storing facial embeddings. It maintains attendance logs with timestamps and provides real-time updates through the web interface. Administrators can manage users and monitor attendance, while students can view their attendance history, making the system comprehensive and user-friendly.

3.3 Non-Functional Requirements

3.3.1 Performance Requirements

The system must operate in real time with minimal latency to ensure smooth user experience. It should efficiently process multiple faces simultaneously without performance degradation. The temporal validation mechanism must work effectively to maintain high accuracy, approximately 96%, while minimizing errors caused by environmental factors.

3.3.2 Safety Requirements

The system ensures that attendance data is stored securely and protected from potential loss or corruption. Error handling mechanisms are implemented to prevent system crashes during continuous operation. Backup mechanisms may be used to maintain data integrity and ensure system reliability.

3.3.3 Security Requirements

AttendNet AI implements authentication and role-based access control to ensure that only authorized users can access sensitive data. User credentials and attendance records are securely stored in the database. The system ensures data confidentiality and prevents unauthorized access or modification.

3.3.4 Software Quality Attributes

The system is designed to be reliable, scalable, and easy to use. It ensures consistent performance under varying conditions and provides high availability. The modular design allows for easy maintenance and future enhancements. Efficiency, usability, and maintainability are key quality attributes considered during system development.

3.4 System Requirements

3.4.1 Software Requirements

The system is developed using Python as the core programming language. Flask is used for backend development, while HTML and CSS are used for frontend design. OpenCV and dlib libraries are utilized for face detection and recognition. Development tools such as Visual Studio Code and Google Colab are used during implementation. A cloud-based database, such as Supabase, is used to store user data and attendance records securely.

3.4.2 Hardware Requirements

AttendNet AI is designed to run on standard computing systems with at least 8 GB RAM and a modern processor such as Intel i5 or AMD Ryzen 5. A webcam is required for capturing real-time video input.

Although GPU support can enhance performance, the system is optimized to function efficiently on CPU-based systems, making it cost-effective and accessible.

3.5 Analysis Model (SDLC Model)

The development of AttendNet AI follows the Waterfall model of the Software Development Life Cycle. In the requirement analysis phase, system requirements are clearly defined. The design phase focuses on developing the system architecture and workflow. During implementation, the system is developed using Python, Flask, and deep learning libraries. The testing phase evaluates system performance, accuracy, and reliability under different conditions. Finally, the system is deployed as a web application, ensuring accessibility and ease of use.

3.6 Limitations of the System

Although AttendNet AI provides an efficient and reliable solution for automated attendance monitoring, certain limitations exist. The system performance is dependent on environmental conditions such as lighting, camera quality, and face visibility. Poor lighting or heavy occlusions may reduce recognition accuracy. Additionally, extreme facial variations such as masks, large accessories, or unusual angles can affect detection and matching.

The system also relies on continuous video input and stable processing, which may introduce slight delays during initial model loading. While the system is optimized for CPU-based execution, performance may vary depending on hardware capabilities. Furthermore, the requirement of internet connectivity for accessing the deployed web application can limit usage in offline environments.

Despite these limitations, the system performs effectively under normal conditions and provides a practical solution for real-world attendance monitoring.

CHAPTER 4

DESIGN AND MODELLING OF SYSTEM

4.1 System Overview

AttendNet AI is designed as a real-time, full-stack face recognition-based attendance monitoring system that integrates computer vision techniques with a web-based application. The system captures live video input through a webcam and processes it to detect faces, extract unique facial features, and identify individuals by comparing them with stored data. A key feature of the system is the incorporation of a temporal validation mechanism, which verifies identity across multiple consecutive frames before marking attendance. This approach significantly improves accuracy and reduces false detections compared to traditional single-frame systems.

The system follows a modular architecture, where each component such as face detection, recognition, database management, and user interface operates independently. This modular design enhances flexibility, maintainability, and scalability, allowing the system to be easily upgraded or extended in the future.

4.2 System Architecture

The architecture of AttendNet AI is based on a client-server model, where the frontend and backend communicate to process user requests and display results. The frontend is developed using HTML and CSS, providing an interactive interface for both administrators and students. The backend is implemented using Flask, which handles core functionalities such as video processing, face recognition, and database interaction.

The system pipeline begins with capturing video input, followed by face detection using HOG-based methods. Detected faces are then processed using a deep learning model from the dlib library to extract 128-dimensional embeddings. These embeddings are compared with stored embeddings using distance metrics such as Euclidean distance to identify individuals. The database stores user information, facial embeddings, and attendance records. The system is deployed on a cloud platform, enabling access through a web browser without requiring local installation, thus ensuring scalability and ease of use.

4.3 System Design Components

Face Detection Module:

This module is responsible for identifying human faces in the video stream. It uses Histogram of Oriented Gradients (HOG) for efficient face detection. Each frame is analyzed to detect facial regions,

and bounding boxes are generated around detected faces. This method is computationally efficient and suitable for real-time applications.

Face Recognition Module:

This module extracts 128-dimensional facial embeddings using a pre-trained deep learning model from the dlib library. These embeddings serve as unique identifiers for each individual. The embeddings are compared with stored data using Euclidean distance to determine identity. The system supports dynamic enrollment, allowing new users to be added without retraining the model.

Temporal Validation Module:

This module is the core innovation of AttendNet AI. Instead of relying on a single frame for recognition, it verifies identity across multiple consecutive frames. A counter is maintained for each detected face, and attendance is marked only when a consistent match is observed. This reduces false positives caused by noise, motion blur, or occlusions.

Database Module:

The database stores user details, facial embeddings, and attendance records. It ensures efficient storage and retrieval of data. Cloud-based solutions such as Supabase are used to provide scalability, reliability, and secure access to stored information.

Web Interface Module:

The frontend provides user-friendly dashboards for administrators and students. Administrators can register users, monitor attendance, and manage records, while students can view their attendance history. The interface is designed to be simple and intuitive for ease of use.

4.4 Data Flow Model (DFD)

The data flow in AttendNet AI starts with capturing video input from the webcam. The captured frames are passed to the face detection module, which identifies faces in each frame. The detected faces are then processed by the recognition module to extract embeddings and compare them with stored data.

The temporal validation module ensures that the detected identity is consistent across multiple frames before confirming attendance. Once validated, the attendance data is recorded in the database along with a timestamp. Finally, the updated attendance information is displayed on the web interface for administrators and students. This continuous flow ensures efficient and real-time processing of attendance data.

4.5 UML Diagram (Use Case Description)

The system involves two primary actors: Administrator and Student. The Administrator has privileges to perform operations such as user registration, attendance monitoring, and system management. The Student can access the system to view their attendance records.

The use case diagram represents the interaction between these actors and the system. Key functionalities include login authentication, face registration, real-time attendance marking, and report viewing. This model helps in understanding system behavior and user interactions clearly.

4.6 Design Considerations

The design of AttendNet AI focuses on achieving high accuracy, efficiency, and scalability. The use of pre-trained deep learning models reduces computational complexity and eliminates the need for retraining. The temporal validation mechanism enhances reliability by ensuring consistent recognition.

The system is optimized to run on standard CPU-based hardware, making it cost-effective and accessible. Web-based deployment ensures that the system can be accessed from any device with a browser. Security measures such as authentication, role-based access control, and data protection are implemented to ensure safe and secure operation. Additionally, the modular architecture allows easy maintenance and future enhancements.

CHAPTER 5

ALGORITHM AND METHODS

5.1 Overview of Methodology

The AttendNet AI system follows a structured and systematic methodology for real-time face recognition and automated attendance marking. The process begins with capturing live video input through a webcam, which is then processed frame by frame. Each frame undergoes face detection, followed by feature extraction using deep learning techniques. The extracted features are compared with stored data to identify individuals.

A key enhancement in this system is the integration of a temporal validation mechanism, which verifies identity across multiple consecutive frames before confirming attendance. This ensures that recognition is not based on a single instance, thereby improving accuracy and reducing false detections. The overall methodology combines computer vision, deep learning, and web technologies to provide an efficient and scalable solution.

5.2 Face Detection Algorithm

Face detection in AttendNet AI is performed using the Histogram of Oriented Gradients (HOG) method, which analyzes gradient orientation patterns to detect facial structures. The video stream is divided into frames, and each frame is processed to locate potential face regions. The HOG algorithm extracts edge and gradient features, which are then used to identify faces by comparing them with learned patterns.

Bounding boxes are generated around detected faces, which helps in isolating the region of interest for further processing. This method is computationally efficient and suitable for real-time applications, especially on CPU-based systems. Its robustness to minor variations in lighting and pose makes it an ideal choice for the system.

5.3 Face Recognition Method

The face recognition process is based on a deep learning approach that converts detected faces into numerical representations known as embeddings. AttendNet AI uses a pre-trained model from the dlib library to generate 128-dimensional embeddings for each detected face. These embeddings capture unique facial features and act as identifiers for individuals.

The generated embeddings are compared with stored embeddings in the database using Euclidean distance. If the calculated distance is below a predefined threshold, the system identifies the face as a

match. This method allows efficient comparison and supports dynamic enrollment, enabling new users to be added without retraining the model. It ensures both accuracy and scalability in real-time applications.

5.4 Temporal Validation Algorithm

The temporal validation mechanism is one of the most important components of AttendNet AI. Instead of relying on single-frame recognition, the system tracks recognition results across multiple consecutive frames. For each detected identity, a counter is maintained to record how many times the face has been consistently recognized.

Attendance is marked only when the identity is verified continuously for a predefined number of frames. This approach minimizes errors caused by temporary disturbances such as motion blur, lighting fluctuations, or partial occlusion. By ensuring consistency over time, the system significantly improves reliability and reduces false positives, making it suitable for real-world environments.

5.5 Attendance Marking Logic

Once a face is successfully validated through the temporal mechanism, the system proceeds to mark attendance. The attendance is recorded in the database along with a timestamp, ensuring accurate tracking of entry time. To prevent duplicate entries, the system checks whether attendance has already been recorded for the individual within the same session.

The attendance data is then updated in real time and displayed on the web interface. Administrators can monitor attendance records, while students can view their attendance status. This logic ensures data consistency, avoids redundancy, and provides real-time accessibility of attendance information.

5.6 Overall Algorithm (Step-wise)

The complete working of the system can be summarized in the following steps:

1. Capture live video stream from the webcam.
2. Extract frames continuously from the video stream.
3. Detect faces in each frame using the HOG-based detection algorithm.
4. Extract 128-dimensional embeddings for each detected face.
5. Compare embeddings with stored data using Euclidean distance.
6. If a match is found, update the frame count for that identity.
7. Apply temporal validation by checking consistency across multiple frames.
8. If validation criteria are satisfied, confirm identity.
9. Mark attendance and store the data with timestamp in the database.

10. Display recognition results and attendance status on the web interface in real time.

This step-by-step process ensures efficient and accurate functioning of the system.

5.7 Methods and Tools Used

The implementation of AttendNet AI utilizes a combination of programming languages, libraries, and frameworks. Python is used as the core programming language due to its simplicity and extensive support for machine learning and computer vision. OpenCV is used for video capture and image processing tasks, while dlib is used for face detection and extraction of facial embeddings.

Flask is used as the backend framework to handle server-side operations and communication between modules. HTML and CSS are used for designing the frontend interface, providing a user-friendly experience. A cloud-based database such as Supabase is used to store user information and attendance records securely. The system is deployed online, ensuring accessibility, scalability, and ease of use across different platforms.

CHAPTER 6

Implementation and Testing

6.1 Implementation Overview

The implementation of AttendNet AI involves the integration of computer vision, deep learning, and web technologies to develop a real-time face recognition-based attendance system. The system is built using Python as the core programming language due to its flexibility and strong support for machine learning libraries. OpenCV is used for real-time video processing, while dlib is used for face detection and feature extraction.

A Flask-based backend is implemented to handle application logic, process requests, and manage communication between system components. The frontend is developed using HTML and CSS to provide a simple and user-friendly interface for both administrators and students. The system is deployed as a web application, enabling users to access it through a browser without requiring installation. This implementation ensures portability, scalability, and ease of use.

6.2 Data Acquisition and Preprocessing

The implementation process of AttendNet AI begins with the data acquisition stage, where facial images of students are collected using a webcam or camera device. During the registration phase, multiple images of each student are captured to account for slight variations in facial expressions, head positions, and lighting conditions. These images are stored in a structured format, typically organized into folders or databases based on user identity. Proper organization of the dataset is essential for efficient retrieval and processing during the recognition phase.

Once the images are collected, preprocessing is performed to standardize and improve the quality of the input data. Initially, the images are resized to a uniform resolution to ensure consistency across the dataset and to reduce computational complexity. Normalization is then applied to adjust pixel intensity values, which helps in minimizing the impact of lighting variations and improves the robustness of the model.

Face alignment is another important step in preprocessing, where facial landmarks such as eyes, nose, and mouth are used to align faces in a consistent orientation. This ensures that all facial images are positioned similarly, which improves the accuracy of feature extraction. Additionally, face regions are extracted from the images using detection techniques, removing unnecessary background information that does not contribute to recognition. This step reduces noise and focuses only on relevant facial features.

After preprocessing, the cleaned and aligned face images are passed through a deep learning model to generate 128-dimensional facial embeddings. These embeddings act as compact and unique representations of each individual's face. The generated embeddings are then stored in the database along with corresponding user details.

Proper data acquisition and preprocessing play a critical role in the overall performance of the system. High-quality and well-preprocessed data ensures accurate recognition, reduces false detections, and improves system reliability under varying real-world conditions such as changes in lighting, pose, and facial expressions.

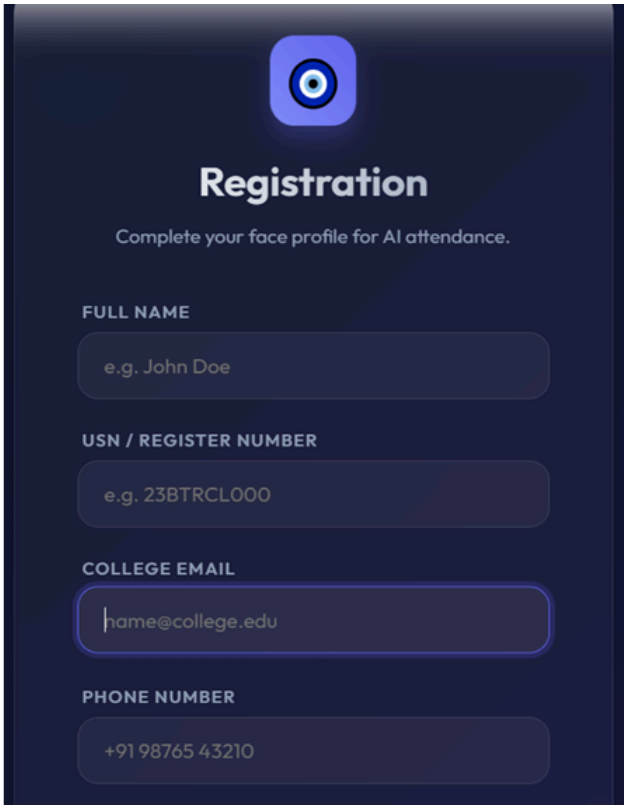

Figure 6.1: User registration interface for dataset creation

6.3 Model Implementation

AttendNet AI utilizes a pre-trained deep learning model from the dlib library to generate 128-dimensional facial embeddings. This model is based on a deep residual network architecture, which provides high accuracy without requiring retraining. During the enrollment phase, embeddings are generated for each user and stored in the database.

During recognition, embeddings extracted from live video frames are compared with stored embeddings using Euclidean distance. If the similarity between embeddings meets a predefined threshold, the identity is confirmed. This approach allows fast and efficient recognition while supporting dynamic addition of new users, making the system scalable and adaptable.

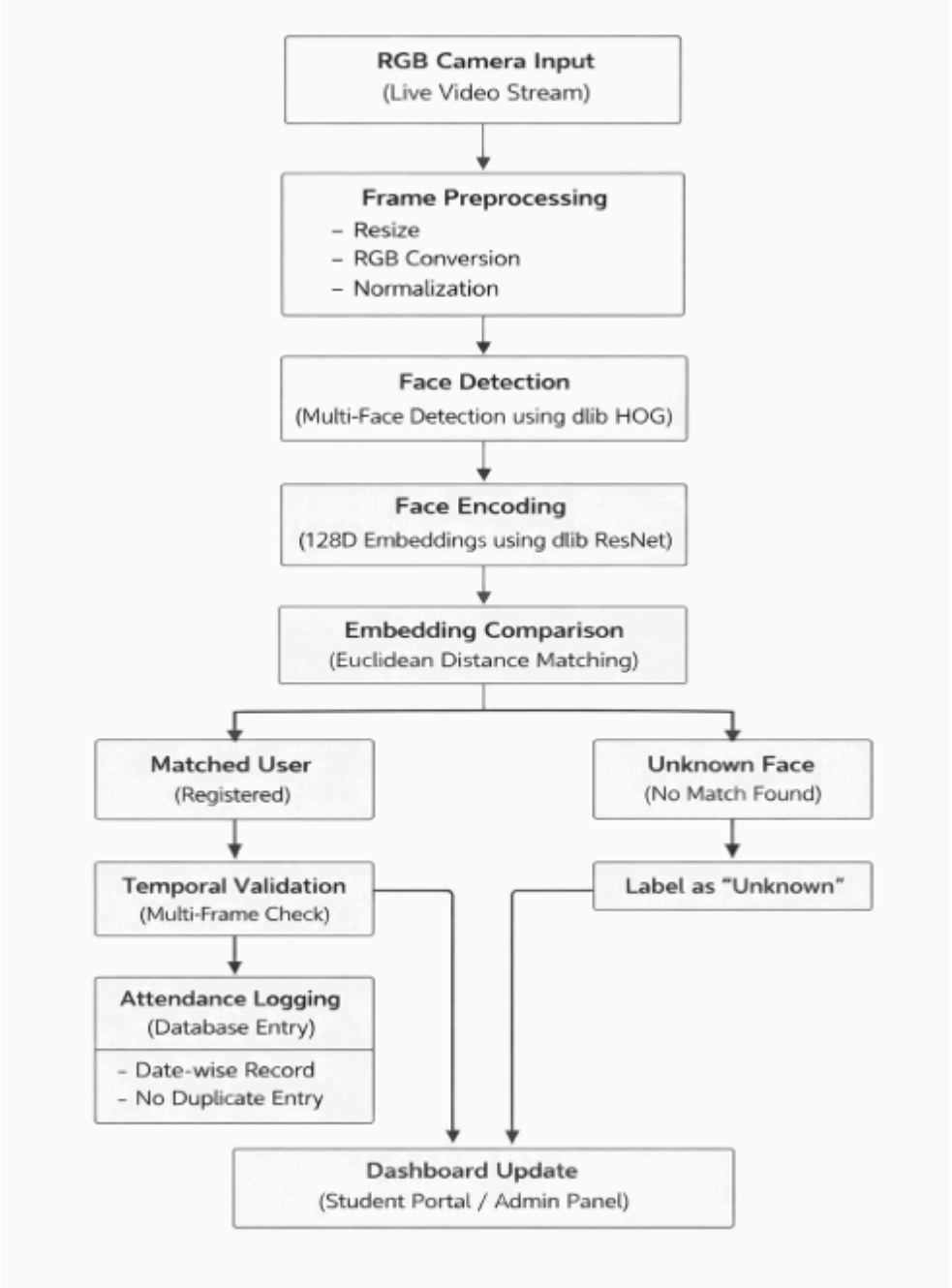

Figure 6.2: Architecture diagram of AttendNet AI system

6.4 SYSTEM INTEGRATION

The AttendNet AI system integrates multiple modules such as face detection, face recognition, temporal validation, database management, and the web interface into a unified framework. Each

module performs a specific function, and together they ensure the smooth operation of the overall system. The integration process begins with capturing video frames from the webcam, which are continuously processed in real time. These frames are passed to the face detection module, which identifies and extracts face regions from each frame.

The detected faces are then forwarded to the recognition module, where facial embeddings are generated and compared with stored data. The recognition results are not immediately used for attendance marking; instead, they are sent to the temporal validation module. This module verifies the consistency of recognition across multiple frames by maintaining a count for each detected identity. Only when the identity is confirmed over a predefined number of frames is it considered valid.

6.5 TESTING METHODOLOGY

The testing of AttendNet AI is carried out using a combination of systematic testing approaches to ensure high accuracy, reliability, and robustness of the system. The goal of testing is to validate that each component of the system performs as expected individually and also works efficiently when integrated into the complete pipeline.

Initially, **unit testing** is performed on individual modules such as face detection, feature extraction, and face recognition. Each module is tested independently using sample inputs to verify its functionality and correctness. For example, the face detection module is tested to ensure it correctly identifies faces in different conditions, while the recognition module is evaluated to confirm accurate embedding generation and matching. This step helps in identifying and fixing errors at an early stage.

After unit testing, **integration testing** is conducted to ensure that all modules interact seamlessly when combined. This includes verifying the correct flow of data between face detection, embedding extraction, temporal validation, and database storage modules. Integration testing ensures that there are no inconsistencies, delays, or communication failures between components, and that the system functions as a unified whole.

Real-time testing is a crucial part of the evaluation process, where the system is tested using live video streams in practical environments. Multiple users are tested simultaneously to evaluate the system's capability to handle real-time multi-face recognition. This testing helps in analyzing system performance under real-world conditions such as classroom settings, where factors like movement, crowding, and varying distances are present.

Additionally, the system is tested under different environmental conditions, including varying lighting levels, different face orientations, and motion scenarios. These tests ensure that the system remains robust and adaptable even when conditions are not ideal. The temporal validation mechanism is also

evaluated during this phase to confirm that it effectively reduces false positives and improves consistency.

Overall, the combination of unit testing, integration testing, and real-time testing ensures that AttendNet AI performs reliably, maintains high accuracy, and operates efficiently in real-world scenarios.

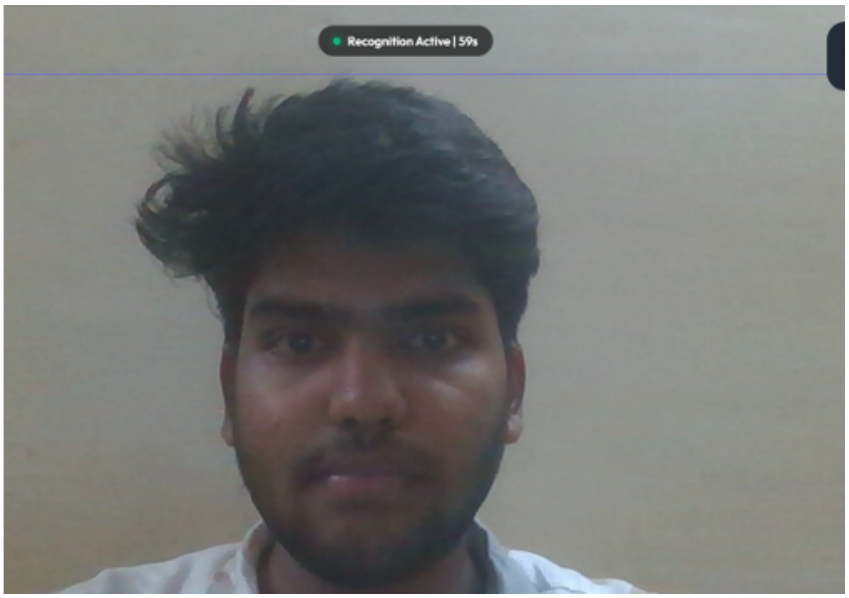

Figure 6.3: Real-time face recognition during attendance marking

6.6 PERFORMANCE EVALUATION

The performance of AttendNet AI is evaluated using key metrics such as accuracy, response time, reliability, and overall system efficiency. These metrics help in assessing how effectively the system performs real-time face recognition and attendance marking under practical conditions. The system achieves an accuracy of approximately 96%, indicating that it correctly identifies the majority of individuals with minimal errors. This high accuracy demonstrates the effectiveness of the deep learning-based embedding approach used for face recognition.

In addition to accuracy, the system's reliability is evaluated by analyzing its ability to maintain consistent performance over time. The temporal validation mechanism plays a crucial role in enhancing reliability by verifying identity across multiple consecutive frames before marking attendance. This significantly reduces false positives and ensures that attendance is recorded only when the system is confident about the identity, thereby improving overall consistency and trustworthiness of the system.

Response time is another important factor in performance evaluation. AttendNet AI processes video frames in real time with minimal delay, ensuring smooth and continuous operation. This makes the system suitable for classroom environments where multiple students may be present simultaneously and quick processing is required. The system is optimized to run efficiently on CPU-based hardware, ensuring that real-time performance is achieved without the need for high-end computational resources.

Furthermore, the system's capability to handle multiple faces simultaneously is also evaluated. AttendNet AI efficiently detects and recognizes multiple individuals in a single frame without significant performance degradation. This demonstrates its scalability and ability to function effectively in crowded environments. Overall, the evaluation confirms that the system provides a balanced combination of accuracy, speed, and reliability, making it a practical solution for real-world attendance monitoring applications.

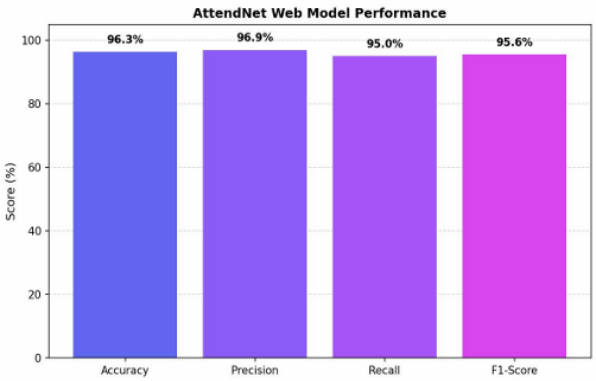

Figure 6.4: Performance metrics graph showing accuracy, precision, recall and F1-score

6.7 RESULTS AND OBSERVATIONS

The results demonstrate that AttendNet AI effectively automates the attendance marking process using face recognition. The system accurately detects and recognizes multiple faces in real time and records attendance with high reliability.

The implementation of temporal validation significantly improves performance compared to traditional single-frame systems by reducing errors caused by noise or temporary disturbances. The web interface provides a convenient platform for administrators to monitor attendance and for students to access their records. Overall, the system performs efficiently under real-world conditions and meets the project objectives.

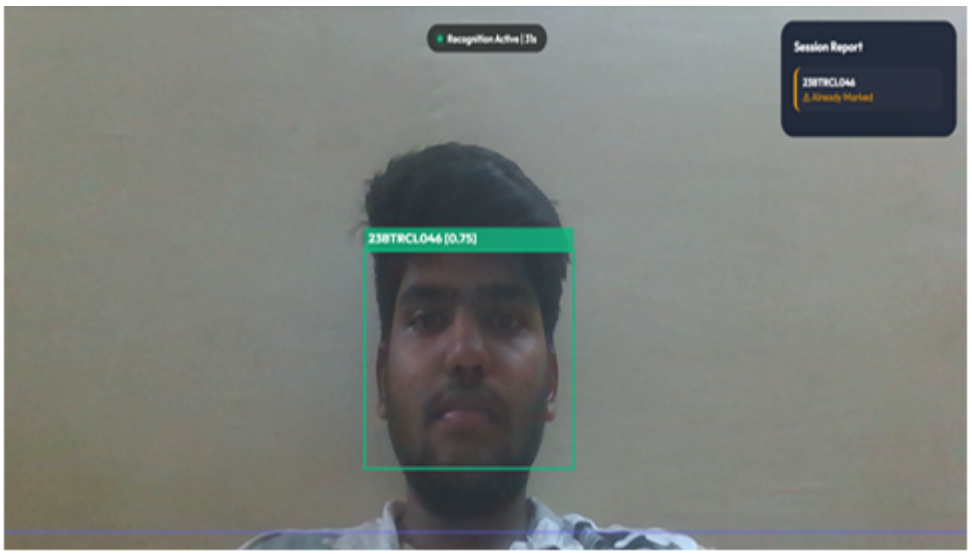

Figure 6.5: Attendance records displayed in system dashboard

6.8 DEPLOYMENT

The AttendNet AI system is deployed as a web application using cloud platforms, making it accessible from any device with a web browser. This deployment approach ensures high scalability and eliminates the need for specialized hardware or local installation, allowing users to access the system easily from different locations. By hosting the application online, the system becomes platform-independent and can be used across various devices such as laptops, tablets, and smartphones.

The application is specifically deployed using **Hugging Face Spaces**, which provides a reliable and user-friendly cloud environment for hosting machine learning and AI-based applications. This platform enables seamless deployment, easy integration, and continuous accessibility of the system through a public URL. It also supports efficient execution of the backend processes, ensuring smooth real-time performance.

Users can access the system online, and all attendance data is securely stored in a cloud-based database. This ensures data integrity, security, and easy retrieval of records whenever required. The use of cloud storage also allows centralized data management, which is beneficial for institutions handling large numbers of students.

Additionally, cloud deployment enables easy updates and maintenance of the system without affecting end users. Any improvements or bug fixes can be implemented on the server side and instantly reflected in the application. This ensures long-term usability and adaptability of the system. Overall, the deployment strategy makes AttendNet AI a practical, scalable, and efficient solution for real-world implementation in educational institutions.

CHAPTER 7

Evaluation

7.1 Evaluation Overview

The evaluation of AttendNet AI is carried out to analyze the system's performance, accuracy, and robustness in real-world scenarios. The system is tested in a classroom-like environment where multiple users are present simultaneously, and various real-life conditions such as lighting variations, facial movements, and partial occlusions are considered. The primary objective of this evaluation is to assess how effectively the system performs real-time face recognition and attendance marking while maintaining consistency and minimizing errors.

In addition to accuracy, the evaluation also focuses on system stability and reliability during continuous operation. The system is observed over extended durations to ensure that performance does not degrade over time. Special emphasis is given to the temporal validation mechanism, which ensures that recognition is consistent across multiple frames rather than relying on a single detection. This helps in reducing errors caused by noise, motion blur, or temporary misdetections. The evaluation provides valuable insights into how well the system can be deployed in practical environments such as classrooms and training centers.

7.2 Evaluation Metrics

To evaluate the performance of the system, standard classification metrics are used. These include True Positive (TP), False Positive (FP), False Negative (FN), and True Negative (TN). True Positive represents correctly identified users, False Positive indicates incorrect identification, False Negative represents missed detections, and True Negative indicates correct rejection of unknown individuals.

Based on these values, important performance metrics such as accuracy, precision, recall, and F1-score are calculated. Accuracy measures the overall correctness of the system, precision indicates how many detected faces are actually correct, recall measures how many actual faces are successfully detected, and F1-score provides a balanced measure of precision and recall.

In addition to these metrics, response time and system latency are also considered to evaluate real-time performance. These metrics provide a comprehensive understanding of system behavior and help in identifying areas for improvement while ensuring reliable operation in real-world conditions.

7.3 Performance Results

The experimental results demonstrate that AttendNet AI achieves high performance in real-time face recognition tasks. The system records an accuracy of approximately 96.3%, a precision of 96.9%, a recall of 95.0%, and an F1-score of 95.6%. These values indicate that the system is highly effective in identifying individuals correctly while minimizing incorrect detections.

The high precision value shows that the system rarely marks attendance for the wrong person, which is critical for maintaining data integrity. The recall value indicates that most registered users are successfully recognized, even under varying environmental conditions. The F1-score reflects a balanced performance, confirming that the system maintains both accuracy and reliability.

These results clearly validate the effectiveness of the proposed approach and demonstrate that the system is suitable for real-time deployment in practical scenarios.

7.4 Impact of Temporal Validation

The temporal validation mechanism is a key innovation in AttendNet AI and plays a significant role in improving system performance. Unlike traditional systems that rely on single-frame recognition, this system verifies identity across multiple consecutive frames before marking attendance.

This approach reduces false positives caused by temporary disturbances such as motion blur, lighting changes, or partial occlusions. It ensures that only consistent and repeated detections are considered valid, thereby increasing system reliability.

As a result, the system becomes more stable and dependable, especially in dynamic environments where users may move frequently. The evaluation clearly shows that temporal validation enhances recognition accuracy, improves consistency, and significantly reduces errors in real-time operation.

7.5 Real-Time Multi-Face Evaluation

The system is tested in scenarios where multiple users are present simultaneously in front of the camera. AttendNet AI successfully detects and recognizes multiple faces in real time without significant performance degradation.

The system maintains stable processing even when handling multiple faces, demonstrating its scalability and efficiency. The ability to track and validate multiple identities simultaneously is an important feature for classroom environments where several students are present.

After the initial model loading phase, the system operates smoothly with minimal delay, ensuring continuous real-time performance. This evaluation confirms that the system can efficiently handle multi-user scenarios without compromising accuracy or speed.

7.6 System Efficiency

The evaluation results indicate that AttendNet AI operates efficiently on CPU-based systems without requiring GPU acceleration. This makes the system cost-effective and accessible for institutions with limited computational resources.

Although there is a slight delay during the initial startup due to model loading, the system performs smoothly during continuous operation. Efficient resource utilization ensures that the system maintains real-time performance without excessive hardware requirements.

This highlights the practicality and scalability of the system, making it suitable for deployment in various environments without the need for high-end infrastructure.

7.7 Observations and Analysis

During testing, the system is exposed to various real-world conditions such as changes in lighting, different facial expressions, and subject movement. Despite these challenges, the system maintains consistent performance due to the implementation of temporal validation.

The system effectively rejects unknown faces, ensuring that only registered users are marked present. It also prevents duplicate attendance entries within the same session, improving data accuracy and reliability.

These observations confirm that the system is robust, reliable, and capable of handling real-world scenarios effectively, making it suitable for practical deployment.

7.8 Comparison with Existing Methods

The performance of AttendNet AI is compared with both traditional and advanced face recognition methods. Conventional methods such as Haar Cascade combined with Local Binary Patterns Histogram (LBPH) show lower accuracy and are highly sensitive to environmental variations.

Advanced models such as DeepFace and FaceNet achieve high accuracy but require significant computational resources and large datasets, making them less suitable for real-time applications on standard hardware.

In contrast, AttendNet AI achieves comparable accuracy while maintaining efficiency and scalability. The inclusion of temporal validation further enhances system reliability, making it more suitable for real-time attendance monitoring.

This comparison highlights the effectiveness of AttendNet AI in balancing accuracy, efficiency, and practicality, making it a strong candidate for real-world deployment.

Method	Accuracy (%)	Remarks
Haar Cascade + LBPH	78.3	Sensitive to lighting and pose variations
DeepFace	96.4	High accuracy, requires heavy computation
FaceNet	97.2	Requires large-scale training data
Proposed AttendNet AI	96.3	Real-time, CPU-based, with temporal validation

Table 7.1: Comparison with Existing Face Recognition Methods

CHAPTER 8

Publication

8.1 Publication Details

As part of this project, the developed system titled *"AttendNet: A Real-Time Deep Learning Framework for Multi-Face Recognition and Automated Attendance Monitoring"* has been successfully transformed into a conference paper and published on an open-access research repository. This publication represents the practical and research-oriented outcome of the project, demonstrating its relevance in the field of computer vision and artificial intelligence.

The paper is published on Zenodo, a widely recognized open-access repository that ensures global accessibility and visibility of research work. Publishing on such a platform allows researchers, students, and professionals to access and reference the work easily. The publication is assigned a Digital Object Identifier (DOI), which provides a permanent and unique link for accessing, sharing, and citing the paper in future research.

The DOI of the published paper is:
https://doi.org/10.5281/zenodo.19654074

In addition to the preprint publication, the paper has also been submitted to peer-reviewed journals for further validation and recognition. The review process is currently ongoing and typically takes around 1 to 1.5 months, depending on the journal's review cycle. This step enhances the credibility of the research and provides an opportunity for formal academic publication.

8.2 Abstract of the Publication

The conference paper presents a real-time face recognition-based attendance system designed to detect and identify multiple individuals simultaneously in a live video stream. The system leverages deep learning-based facial embeddings to generate unique feature representations for each individual. These embeddings are compared using distance-based matching techniques, enabling accurate and efficient identification without the need for retraining the model.

To improve system reliability and reduce errors, a temporal validation mechanism is incorporated, which verifies identity across multiple consecutive frames before marking attendance. This approach significantly reduces false positives and enhances consistency in dynamic environments.

The proposed system achieves an accuracy of approximately 96% in real-time scenarios, demonstrating its effectiveness and practicality. The system is suitable for applications such as

classroom attendance monitoring and can be extended to other domains requiring automated identification and tracking. Overall, the research highlights the integration of deep learning and real-time processing to develop a scalable and reliable attendance management solution.

8.3 Publication Proof

The proof of publication is provided below:

AttendNet: A Real-Time Deep Learning Framework for Multi-Face Recognition and Automated Attendance Monitoring

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Abstract-Automated face recognition attendance systems offer a contactless alternative to biometric and hardware-based roll-call and biometric attendance systems. Nevertheless, the majority of current solutions are based on single-frame recognition, making them unstable in dynamic scenarios like a classroom where objects are in constant motion, and the lighting is not constant.

In this paper, the author will introduce AttendNet AI, a temporal-sensitive real-time face recognition system that can be used to monitor attendance automatically. The system proposed is an identity matching model that uses a distance-based similarity measure to match identities with a pre-trained deep metric model that generates facial embeddings. To increase the reliability, a multi-frame temporal validation mechanism is added, where the recognition is reliable between consecutive frames before marking attendance.

The system is deployed as a full-stack application with built-in web interface, secure administration authentication, student portal and database based attendance logging. Experimental analysis shows good performance with an accuracy of 96.3, a precision of 96.9, a recall of 95.0 and an F1-score of 95.6.

The findings reveal the enhanced resilience to motion, occlusion, and transient detections, and the system can

RFID based solutions which are able to automate attendance, they need specific hardware, physical interaction and may not be scalable in large classroom settings.

As AI and computer vision continue to rapidly evolve, face recognition has become one of the potential alternatives to automated attendance systems that is contactless. Deep learning techniques have also proven to be very accurate at facial recognition by deriving feature representations that are discriminative. Most current systems, however, are based on single frame recognition, which is not reliable in real world scenarios where the subject moves a lot, the lighting changes and faces are partially blocked. Such limitations tend to cause false detection and wrong attendance marking.

In order to overcome these issues, this paper presents AttendNet AI, a temporal-aware real-time face recognition system, which combines multi-frame validation with embedding-based recognition. The method suggested will make sure a face is always detected in successive frames before identity is verified and therefore false positives due to temporary

Figure 8.1: First page of the published conference paper

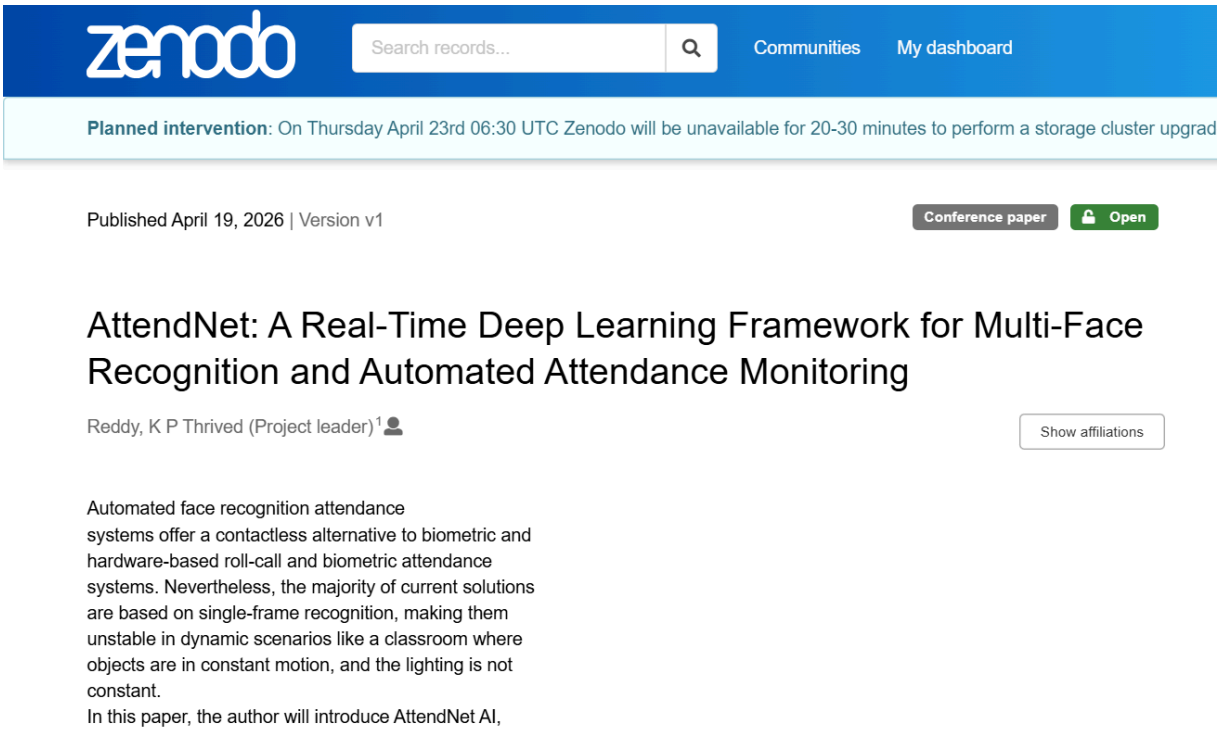

Figure 8.2: Zenodo publication page showing conference paper details and DOI

8.4 Significance of Publication

The publication of this research work provides several advantages:

- Ensures global accessibility of the research work
- Provides a permanent DOI for citation and reference
- Demonstrates practical implementation of AI techniques
- Enhances academic and research profile
- Enables early dissemination of research findings

8.5 Journal Submission Status

Along with the open-access publication, the research paper has been submitted to relevant peer-reviewed journals for formal evaluation. The paper is currently under review, and the review process is expected to take approximately 1 to 1.5 months.

This approach follows the standard research workflow, where a preprint is published for early access, followed by submission to journals for peer-reviewed validation and academic recognition.

CHAPTER 9

8.1 CONCLUSION

The AttendNet AI system presents an effective and innovative solution for automated attendance monitoring by leveraging real-time face recognition and deep learning techniques. It successfully addresses the major limitations of traditional attendance systems such as manual roll calls, proxy attendance, and dependency on hardware-based biometric devices. By adopting a contactless approach, the system enhances convenience, hygiene, and operational efficiency, making it highly suitable for modern educational environments.

The integration of deep learning-based facial embeddings enables accurate and reliable identification of individuals by capturing unique facial features. A significant contribution of this system is the implementation of the temporal validation mechanism, which verifies identity across multiple consecutive frames before confirming attendance. This approach reduces false positives caused by temporary disturbances such as motion blur, lighting variations, or partial occlusions, thereby improving system robustness and consistency.

The system demonstrates strong performance under real-world conditions, achieving an accuracy of approximately 96% along with balanced precision and recall values. It is capable of detecting and recognizing multiple faces simultaneously in a live video stream without significant performance degradation. This makes it highly suitable for classroom environments where multiple students are present at the same time.

Furthermore, the implementation of AttendNet AI as a full-stack web application enhances its usability and accessibility. Features such as user registration, real-time attendance monitoring, secure database management, and role-based access for administrators and students make the system comprehensive and user-friendly. The ability to operate efficiently on standard CPU-based systems without requiring specialized hardware makes it cost-effective and scalable for widespread adoption.

Overall, AttendNet AI successfully combines computer vision, deep learning, and web technologies to deliver a robust, scalable, and practical attendance management system. The project highlights the potential of intelligent automation in improving administrative processes and demonstrates how advanced technologies can be applied to solve real-world problems effectively. It also provides a strong foundation for future research and development in automated recognition systems.

8.2 FUTURE SCOPE

Although AttendNet AI performs effectively in real-time attendance monitoring, there are several opportunities for further enhancement and expansion. One potential improvement is the integration of lightweight deep learning models such as MobileNet or EfficientNet, which can reduce computational requirements and improve processing speed. This would enable deployment on low-power devices and improve performance in resource-constrained environments.

Advanced preprocessing techniques and data augmentation methods can also be incorporated to improve robustness under challenging conditions such as low lighting, heavy occlusions, and extreme facial variations. Additionally, the system can be enhanced by incorporating adaptive thresholding techniques for better recognition accuracy across diverse datasets.

The system can be extended by integrating cloud-based infrastructure to support large-scale deployment across multiple institutions. This would enable centralized attendance management, real-time synchronization, and easier data analysis. Cloud integration can also improve system reliability and allow remote access from different locations.

Another important area of future development is the integration of multimodal authentication techniques. Combining face recognition with additional methods such as voice recognition, fingerprint verification, or ID-based authentication can further enhance system security and reduce the chances of spoofing or unauthorized access.

Future work may also focus on optimizing the system for edge devices such as smartphones, embedded systems, or IoT-based platforms. This would allow attendance monitoring in remote or offline environments where internet connectivity is limited.

In addition, improvements in user interface design and the development of advanced analytics dashboards can provide deeper insights into attendance patterns, student behavior, and performance trends. These features can assist institutions in decision-making and resource management.

Overall, these enhancements will significantly improve the efficiency, scalability, and applicability of AttendNet AI, making it a more powerful and versatile solution for real-world attendance monitoring systems.

CHAPTER 10

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